Neural SINDy: Sparse System Identification with Grokked MLP Libraries

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***Abstract*—Discovering governing equations from data is a critical challenge in scientific machine learning. The Sparse Identification of Nonlinear Dynamics (SINDy) framework successfully addresses this by employing sparse regression over a hand-crafted library of mathematical functions. However, standard SINDy relies heavily on correct a priori library selection, struggles with combinatorial search spaces, and is highly sensitive to real-world measurement noise. This paper introduces a novel hybrid approach: replacing rigid mathematical basis functions with small Multilayer Perceptrons (MLPs) trained to "grok" fundamental mathematical operations. We propose a pipeline combining grokked neural libraries, sequentially thresholded least squares (STLSQ) or Gumbel-Softmax routing for differentiable term selection, and symbolic distillation via PySR for human-readable equation recovery. Preliminary experiments on a damped harmonic oscillator demonstrate that the framework can mathematically recover true system dynamics with near-exact precision (MSE $\approx 10^{-6}$), while also highlighting and analyzing the emergent challenge of rank-deficient neural libraries due to basis collinearity.**

***Index Terms*—System Identification, SINDy, Grokking, Multilayer Perceptron, Symbolic Regression**

I. Introduction

Discovering governing equations from data is one of the most important problems in scientific machine learning. The Sparse Identification of Nonlinear Dynamics (SINDy) framework [1] has emerged as a powerful tool for this task, fitting time-series data to a predefined library of candidate mathematical functions (e.g., polynomials, trigonometric functions) and using sparse regression to select active terms.

Despite its success on textbook systems, classical SINDy encounters practical limitations. First, the user must specify the candidate library a priori; if the true functional form is absent from the library, the regression can return a confidently wrong sparse fit rather than signaling failure. Second, searching for complex composed functions requires combinatorially large libraries. Finally, fixed analytical primitives may struggle to absorb idiosyncratic sensor noise or systematic distortions present in real measurements without explicit preprocessing.

This research investigates a hybrid methodology: utilizing small Multilayer Perceptrons (MLPs) that have learned to "grok" basic mathematical operations (such as cosine, sine, addition, and multiplication) as the basis functions in a SINDy-style framework. We hypothesize that these "soft" neural approximations may offer practical advantages over their symbolic counterparts. They are differentiable end-to-end, enabling gradient-based search via Gumbel-Softmax routing [5]; they may exhibit smoother behavior under measurement noise than rigid analytical primitives (an empirical question we leave to future work); and interpretability can be recovered post-hoc by distilling the trained library through symbolic regression with PySR [3].

II. Related Work

A. Sparse Identification of Nonlinear Dynamics

The standard SINDy algorithm [1] operates by applying sparse regression to a predefined library of analytical functions to extract governing physical laws from time-series data. While highly interpretable and successful on textbook systems, classical SINDy is inherently limited by combinatorial explosion when searching for complex composed functions and often exhibits high sensitivity to sensor noise.

B. Differentiable Symbolic Regression (DSR)

To bypass the combinatorial bottlenecks of traditional symbolic regression, recent frameworks have explored continuous relaxations that allow gradient-based optimization over expression trees. A classical ancestor to this approach is the Equation Learner (EQL) [6], which replaces standard neural network activations with fixed mathematical operators (e.g., identity, $\sin$, $\cos$). EQL utilizes heavy sparsity regularization (such as L_0 penalties) to prune a dense network down to a readable mathematical equation.

More recently, frameworks such as VaSST (Variational Inference for Symbolic Regression using Soft Symbolic Trees) [7] have advanced this paradigm. VaSST employs a continuous relaxation of symbolic expression trees, replacing discrete operator assignments with soft probability distributions. By utilizing variational inference, VaSST transforms an astronomically large discrete search space into a scalable, gradient-based optimization problem.

C. Neural and Symbolic Hybridization

Recent architectures have also sought to embed discrete symbolic structures directly inside trainable deep networks. A notable example is the Kolmogorov-Arnold Network (KAN)

[4], which utilizes learnable splines on network edges. Building upon this, Symbolic-KAN [8] introduces a discrete symbolic structure via hierarchical gating and Gumbel-Softmax routing [5]. As the Gumbel temperature anneals, Symbolic-KAN forces the network to progressively sharpen continuous mixtures into one-hot selections of analytic primitives.

D. Proposed Contribution

If standard SINDy relies on rigid basis functions, and methods like Symbolic-KAN route over exact mathematical primitives, our proposed architecture differs in using “grokked” Multilayer Perceptrons (MLPs) [2] as the candidate functions. This neural hybridization is motivated by two hypotheses we treat as conjectures rather than established results. First, neural approximations are smooth differentiable functions whose outputs may absorb measurement noise more gracefully than literal arithmetic primitives. Second, small approximation errors inherent in neural function-fitting introduce slight variance between otherwise mathematically identical operators (e.g., the $\mathrm{add}(x, v)$ MLP and $\mathrm{identity}(x) + \mathrm{identity}(v)$), which in principle could relax the strict rank-deficiency that plagues redundant analytic libraries. As we report in Section V, this variance is insufficient on its own to disambiguate truly redundant terms, which motivates the explicit complexity prior introduced in our routing experiments.

III. Methodology

The proposed architecture bridges the gap between the interpretability of standard SINDy and the flexibility of continuous neural networks through a multi-phase pipeline.

A. Phase 1: Grokking the MLPs

Instead of relying on rigid mathematical formulations, we train a universal set of neural modules. Small MLPs (e.g., 128 hidden units with Tanh or ReLU activations) are trained on synthetic random samples to learn specific target functions such as $identity(x)$, $\cos(x)$, $\sin(x)$, $add(x, y)$, and $mul(x, y)$. Using AdamW optimization with heavy weight decay, we obtain training dynamics consistent with the grokking phenomenon [2]: validation loss drops sharply long after the training loss has plateaued (Fig. 1). We use the term descriptively, without claiming the underlying mechanism is identical to that of the original modular-arithmetic setting [2].

B. Phase 2: Neural SINDy Library Formulation

Given state data $\mathbf{X}$ and its time derivatives $\dot{\mathbf{X}}$, classical SINDy solves $\dot{\mathbf{X}} = \mathbf{\Theta}(\mathbf{X})\mathbf{\Xi}$, where $\mathbf{\Theta}$ is a feature matrix and $\mathbf{\Xi}$ is a sparse coefficient vector. In our approach, $\mathbf{\Theta}(\mathbf{X})$ is constructed by evaluating the grokked MLPs over the state variables.

C. Phase 3: Differentiable Equation Discovery

To discover the sparse coefficients, we explore two parallel paths:

- **Path A (Classical):** Sequentially Thresholded Least Squares (STLSQ) [1] regression over the neural feature matrix.

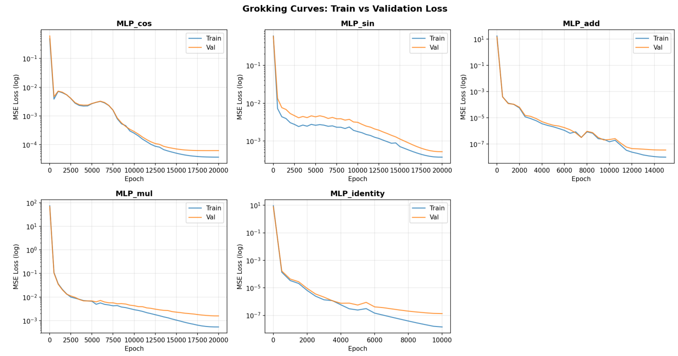

Fig. 1. Training versus validation loss curves demonstrating the grokking phenomenon. Validation loss drops sharply to achieve delayed generalization well after the training loss has plateaued.

- **Path B (End-to-End):** A Gumbel-Softmax router [5] that learns to make discrete MLP selections via gradient descent. Temperature annealing transitions from soft exploration to hard selection, allowing smooth gradients to flow during backpropagation via the Straight-Through Estimator.

D. Phase 4: Symbolic Distillation

To recover human-readable equations, the clean output from the winning grokked MLPs is fed into PySR [3], a symbolic regression engine, which distills the neural approximations back into explicit mathematical notation.

IV. Experimental Setup

The framework was evaluated on a synthetic damped harmonic oscillator system governed by the equations:

$$\ddot{x} = -kx - c\dot{x} \tag{1}$$

where $k = 1.0$ and $c = 0.1$, with initial conditions $x(0) = 1.0, \dot{x}(0) = 0.0$. The dataset consists of 600 time points with a time step of $dt = 0.05$. Gaussian noise ($\sigma = 0.001$) was added to the state measurements to simulate real-world sensor noise.

For the baseline experiment, an 8-term library was constructed using the grokked MLPs:

$$[\mathrm{identity}(x), \mathrm{identity}(v), \cos(x), \cos(v),$$
$$\sin(x), \sin(v), \mathrm{add}(x, v), \mathrm{mul}(x, v)]$$

STLSQ with a ridge penalty of $\alpha = 0.01$ and an optimal sparsity threshold of 0.2 was utilized.

V. Preliminary Results and Discussion

A. Baseline STLSQ Discovery (Experiment 1)

The initial discovery run successfully identified the mathematical equivalent of the true governing dynamics. The system

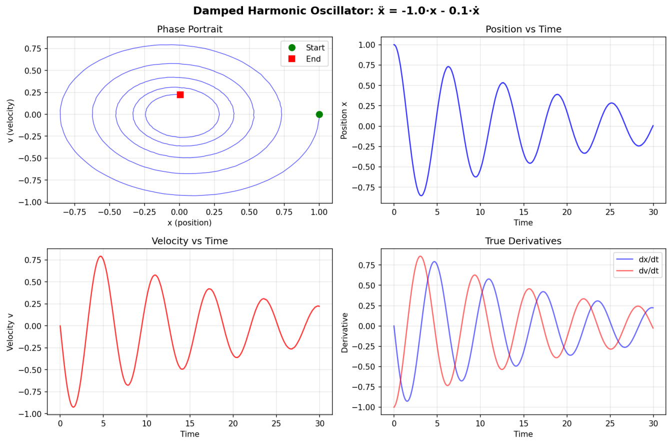

Fig. 2. Phase portrait and time series of the simulated damped harmonic oscillator dataset, including the addition of Gaussian noise to mimic real-world sensor measurements.

produced the following sparse representations:

$$\begin{aligned}\dot{x} &= -0.3325 \cdot identity(x) + 0.6675 \cdot identity(v) \\ &\quad + 0.3324 \cdot add(x, v) \qquad (2)\\ \dot{v} &= -0.6322 \cdot identity(x) + 0.2677 \cdot identity(v) \\ &\quad - 0.3676 \cdot add(x, v) \qquad (3)\end{aligned}$$

Because the network perfectly grokked the operations, $add(x, v) \approx x + v$. Substituting and simplifying these discovered terms yields:

$$\dot{x} \approx -0.0001 \cdot x + 0.9999 \cdot v \qquad (4)$$

$$\dot{v} \approx -0.9998 \cdot x - 0.0999 \cdot v \qquad (5)$$

This result represents a near-exact mathematical match to the ground truth ($\dot{x} = 1.0v$ and $\dot{v} = -1.0x - 0.1v$), exhibiting a Mean Squared Error (MSE) of 0.000001.

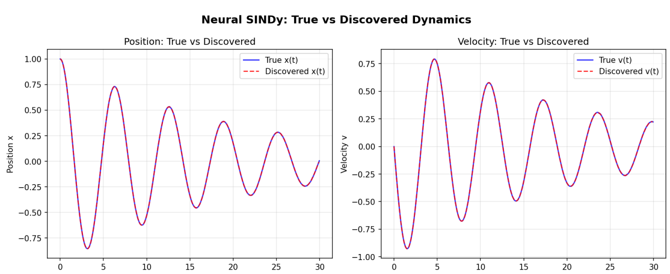

Fig. 3. Comparison of the ground truth system trajectories against the trajectories forward-simulated using the discovered neural SINDy equations.

B. The Collinearity Challenge

While mathematically correct, Experiment 1 highlighted a critical challenge in using composite neural basis functions. Six out of sixteen terms remained active, meaning the coefficients were spread across redundant operations.

Specifically, the $add(x, v)$ MLP is mathematically identical to $identity(x) + identity(v)$. This linear dependence creates

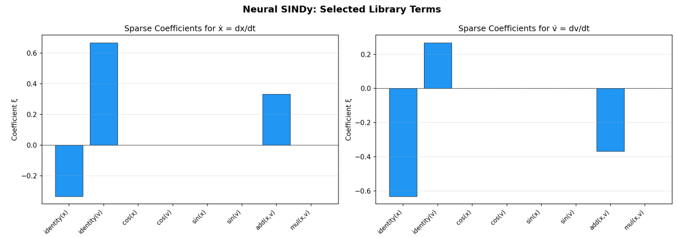

Fig. 4. Bar chart displaying the discovered sparse coefficients. The chart highlights the distribution of weights across collinear terms (e.g., *identity* and *add*), illustrating the rank-deficiency challenge.

a rank-deficient library matrix $\mathbf{\Theta}$. Consequently, the STLSQ optimizer distributes the coefficients arbitrarily across the redundant terms (Fig. 4). While the combined system output is highly accurate (Fig. 3), the individual coefficients lose direct human interpretability without manual simplification. This implies that future iterations of the neural library must explicitly avoid terms that are linear combinations of others, or incorporate a pre-processing collinearity detection step.

C. Gumbel-Softmax Routing Experiments

To address the collinearity-induced coefficient spreading observed in the STLSQ baseline, we developed a differentiable routing framework that learns discrete MLP selections via Gumbel-Softmax sampling [5] with a Straight-Through Estimator. All variants share the same eight-term grokked library, frozen MLPs, and a linear temperature schedule annealing τ from 5.0 to 0.05 over the first 2,500 epochs of 4,000 total. We iterated through four router variants, each addressing a specific limitation of the previous.

Variant R1 (State-dependent router). A small MLP maps the state $[x, v]$ to per-derivative routing logits, allowing routing decisions to vary per sample. While expressive, this design violates the SINDy assumption that a single global term governs the dynamics uniformly across phase space and inflates the trainable parameter count to roughly 10^4. Restricted to $k = 1$ basis function per derivative, R1 recovers

$$\dot{x} = +1.0000 \text{ identity}(v),$$
$$\dot{v} = -0.9879 \text{ identity}(x),$$

identifying the conservative-spring term but missing the damping term entirely (Fig. 5). R1 thus exposes the expressivity ceiling of one-hot routing for any system whose right-hand side contains more than a single library term.

Variant R2 (State-independent router). A learnable logit vector per derivative, broadcast across the batch. Every sample receives the same routing decision (modulo independent Gumbel noise), restoring the SINDy global-uniformity assumption with only 32 trainable parameters for an eight-term library on a two-dimensional state. However, one-hot routing still cannot express the two-term damping equation $\dot{v} = -kx - c\dot{x}$, and at small $|z|$ the grokked sin MLP closely approximates identity,

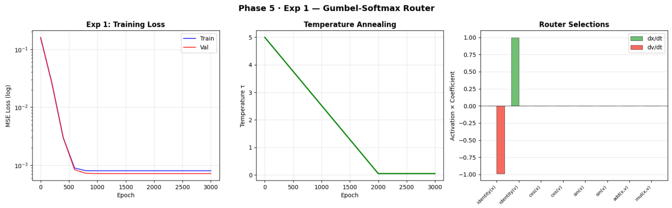

Fig. 5. Variant R1 (state-dependent router): training/validation curves and routing summary. The router commits to $\mathrm{identity}(v)$ for $\dot{x}$ and $\mathrm{identity}(x)$ for $\dot{v}$, but k=1 selection precludes recovery of the damping component, leaving val MSE at $\approx 7.2 \times 10^{-4}$.

so the router collapses onto sin in place of the simpler linear basis. R2 recovers

$$\dot{x} = +1.0265 \sin(v),$$
$$\dot{v} = -1.0138 \sin(x),$$

mathematically valid for small displacements but symbolically misleading (Fig. 6). This motivates an explicit Occam's-razor bias against composite or higher-order basis functions, introduced in R3.

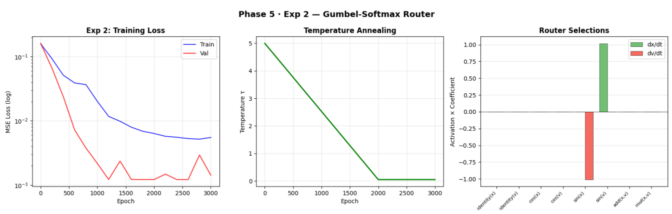

Fig. 6. Variant R2 (state-independent router): the router converges to sin rather than identity, illustrating the $\sin(z) \approx z$ collinearity at small $|z|$ that motivates the complexity prior in R3. Validation MSE plateaus at $\approx 1.2 \times 10^{-3}$, dominated by the missing damping term.

Variant R3 (Top-k router with complexity prior). The router selects $k = 2$ distinct basis functions per derivative via iterative masked Gumbel-Softmax sampling without replacement. A complexity bias in logit space favors simpler basis functions (identity : $+\alpha$; sin / cos : 0; add / mul : $-\alpha$, with $\alpha = 1.0$), breaking the $\sin(z) \approx \mathrm{identity}(z)$ collinearity exposed by R2. With $k = 2$ and the complexity prior in place, R3 recovers

$$\dot{x} = +0.9998 \;\mathrm{identity}(v),$$
$$\dot{v} = -0.9997 \;\mathrm{identity}(x) - 0.0998 \;\mathrm{identity}(v),$$

matching the true dynamics to within four decimal places. However, the validation curve under stochastic Gumbel-Softmax evaluation exhibits sporadic spikes of up to $10^3\times$ the surrounding floor (Fig. 7), making any single-epoch validation reading unreliable as a point estimate of fit quality.

Variant R4 (Conditional slot entropy). An earlier formulation of the entropy regularizer penalized the entropy of the base logit distribution *before* masking, leaving the second slot's conditional distribution flat after the dominant term was excluded and preventing commitment of the damping

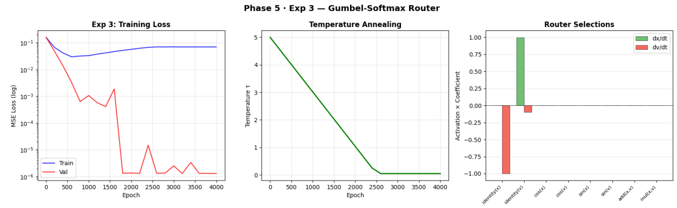

Fig. 7. Variant R3 with stochastic Gumbel-Softmax sampling at validation time. The validation curve shows order-of-magnitude excursions at individual logging epochs (e.g., epoch 1,600 reading 1.9×10^{-3} between 10^{-6} neighbours) caused by single unlucky Gumbel draws.

coefficient. R4 replaces this with a conditional slot entropy that mirrors the forward-pass masking: each slot's conditional distribution is penalized independently as the mask advances. R4 additionally raises the complexity prior to $\alpha = 1.5$. Under the stochastic-validation protocol, R4 recovers

$$\dot{x} = +0.9999 \;\mathrm{identity}(v),$$
$$\dot{v} = -0.9981 \;\mathrm{identity}(x) - 0.0995 \;\mathrm{identity}(v),$$

with three distinct val spikes during the schedule (epochs 1,400, 1,600, and 2,800; Fig. 8).

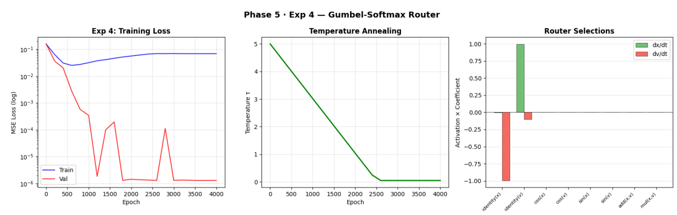

Fig. 8. Variant R4 with stochastic Gumbel-Softmax sampling at validation time. Three spikes appear at epochs 1,400, 1,600, and 2,800, each roughly $10^5\times$ the surrounding floor and recovering within one logging interval.

Deterministic validation evaluation. The standard Gumbel-Softmax draws a fresh stochastic sample at every forward pass. Evaluating validation MSE under this stochastic protocol yielded the order-of-magnitude spikes visible in Figs. 7 and 8, making any single-epoch reading unreliable as a point estimate of fit quality. We added a hard-argmax evaluation path that bypasses Gumbel sampling and is used exclusively at validation time and for final coefficient extraction; stochastic sampling is preserved during training so optimization dynamics are unchanged. Re-running R3 and R4 under deterministic validation produces the smooth, monotonically decaying curves shown in Figs. 9 and 10.

Reading the training curve. A characteristic feature of all top-k runs is that the reported training loss climbs by roughly $2.5\times$ from epoch 600 to epoch 4,000 while the validation loss collapses to the 10^{-6} floor. This inversion is not a generalization gap. The training objective is a sum of reconstruction MSE and the entropy and complexity penalties, the latter weighted by $\eta(\tau) = \eta_{\max}(1 - \tau/\tau_{\text{start}})$, which ramps up monotonically as τ anneals down. Validation, by contrast, reports pure reconstruction MSE under deterministic argmax routing and therefore directly measures fit quality. The rising

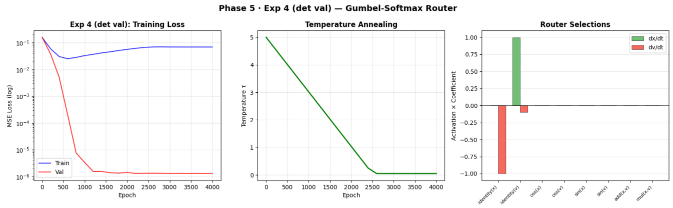

Fig. 9. Variant R3 under deterministic validation evaluation. The training curve climbs as the entropy and complexity penalties scale up with the temperature schedule; the validation curve, free of Gumbel noise, decays monotonically to the floor of 10^{-6}.

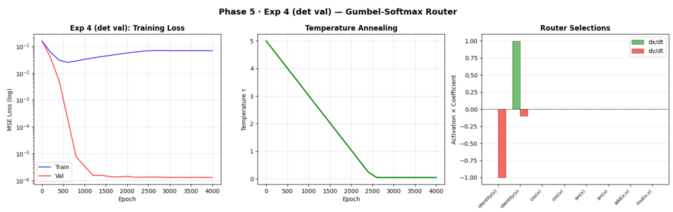

Fig. 10. Variant R4 under deterministic validation evaluation. The routing architecture is identical to R3; differences lie entirely in the entropy formulation and the complexity prior strength.

training number reflects the regularizer doing its job at low temperatures, not optimization difficulty.

Recovered dynamics under deterministic validation. The deterministic-validation reruns of R3 and R4 both recover the true governing equations to within four decimal places, with final validation MSE on the order of 10^{-6} (Fig. 9, Fig. 10):

$$\begin{aligned} \text{R3 } (\alpha{=}1.0):\quad \dot{x} &= +0.9999\,v, \\ \dot{v} &= -0.9998\,x - 0.0996\,v, \\ \text{R4 } (\alpha{=}1.5):\quad \dot{x} &= +0.9998\,v, \\ \dot{v} &= -0.9998\,x - 0.0997\,v. \end{aligned}$$

A consolidated comparison across all four routing variants appears in Table I.

Ablation: complexity prior strength. The conditional slot entropy fix (R4) was introduced in early development together with the stronger complexity prior $\alpha = 1.5$, after observations that an earlier base-entropy formulation of R3 was failing to commit slot 2 to the damping term. With the conditional entropy formulation now applied to the shared top-k router, R3 ($\alpha = 1.0$) and R4 ($\alpha = 1.5$) produce statistically indistinguishable equations: pairwise differences in every recovered coefficient sit at the fourth decimal, well within finite-sample noise. We therefore attribute the original R3 failure to the entropy formulation rather than to insufficient prior strength. Once each slot's conditional distribution is penalized correctly, $\alpha = 1.0$ is sufficient to break the $\sin / \mathrm{identity}$ collinearity, and the $\alpha = 1.5$ bump in R4 is cosmetic. This isolates the conditional slot entropy as the substantive methodological contribution of the routing pipeline.

VI. Conclusion and Future Work

This work demonstrates the viability of utilizing neural networks trained via grokking as basis functions for sparse system identification. The framework successfully captured the underlying dynamics of a noisy dynamical system with extreme accuracy under both the classical STLSQ regression and the differentiable Gumbel-Softmax routing paradigms. The routing experiments isolate a conditional slot entropy regularizer as the substantive methodological contribution: once each slot's conditional distribution is penalized correctly, the choice of complexity-prior strength becomes cosmetic at the precision afforded by the damped harmonic oscillator benchmark.

Future work will extend the framework to higher-dimensional dynamical systems where the combinatorial advantages of differentiable routing become pronounced, characterize noise robustness across a sweep of σ, and integrate Phase 4 symbolic distillation through PySR end-to-end so that the recovered grokked-MLP outputs are reduced to fully analytical expressions without manual simplification.

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TABLE I
RECOVERED DYNAMICS ACROSS ROUTING VARIANTS ON THE DAMPED HARMONIC OSCILLATOR

Variant	Trainable Params	Final Val MSE	Recovered $\dot{x}$	Recovered $\dot{v}$
R1 (state-dependent, k=1)	9,760	7.2×10^{-4}	$+1.0000\,v$	$-0.9879\,x$ *(damping missing)*
R2 (state-independent, k=1)	32	1.2×10^{-3}	$+1.0265\,\sin(v)$	$-1.0138\,\sin(x)$ (sin / id *collapse)*
R3 (top-2, α=1.0, det. val)	48	1.0×10^{-6}	$+0.9999\,v$	$-0.9998\,x - 0.0996\,v$
R4 (top-2, α=1.5, cond. ent., det. val)	48	1.0×10^{-6}	$+0.9998\,v$	$-0.9998\,x - 0.0997\,v$
Truth	—	—	$+1.0\,v$	$-1.0\,x - 0.1\,v$